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A CAPSTONE PROJECT REPORT
ON
SUSPICIOUS ACTIVITY DETECTION IN SURVEILLANCE VIDEO
STREAMS

A Capstone Project Report submitted in partial fulfilment of the requirements
for the Course of

BACHELOR OF TECHNOLOGY
IN
ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

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CERTIFICATE FROM THE SUPERVISOR

This is to certify that the Capstone Project entitled “Suspicious Activity Detection in Surveillance Video Streams” has been carried out by the student under the supervision of the project supervisor and is submitted in partial fulfilment of the requirements for the current semester of the Bachelor of Technology program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

I hereby declare that the Capstone Project Work entitled “Suspicious Activity Detection in Surveillance Video Streams” is the result of my own bonafide efforts. To the best of my knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, datasets, software libraries, computational resources, and AI-assisted tools used during the project have been appropriately acknowledged. I further declare that the data, results, analysis, and findings presented in this report have not been knowingly fabricated, falsified, or manipulated.

Place: Chennai

Date: 02/09/2026

Signature of the Students with Names

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INDIVIDUAL CONTRIBUTION STATEMENT

KOTA PUJITHA (Individual Project)	System analysis, implementation, integration and project demonstration	Real-time detection pipeline and Real-Time Detection & Alert Generation module	Detection, alert-response and integrated workflow verification	Ch. 1, Ch. 3, Ch. 4, Ch. 5, Appendix J	100%
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Total: 100%

ABSTRACT

The Suspicious Activity Detection in Surveillance Video Streams project is designed to provide an intelligent and automated approach for identifying potentially suspicious events in live or recorded surveillance footage. Conventional surveillance depends heavily on human operators continuously watching multiple camera feeds, which can lead to fatigue, delayed response, and missed events. The proposed system uses computer vision techniques to analyse video frames and detect activities that require attention.

The system focuses on the Real-Time Detection & Alert Generation module. Video frames are captured continuously from a camera or surveillance stream, resized and preprocessed, and analysed using an object-detection model and activity rules. Detected people, vehicles, restricted-area entry, prolonged loitering, unusual movement, crowd formation, or other configured events can be monitored over time. When a suspicious condition satisfies the defined threshold, the system generates an alert containing the event type, confidence, time, and camera information.

The proposed solution aims to reduce the monitoring workload and improve the speed of security response. A simple monitoring interface can display live detections, bounding boxes, event labels, confidence values, alert status, and event history. The system is intended as a decision-support tool rather than a replacement for trained security personnel. Future improvements can include stronger activity-recognition models, multi-camera tracking, edge deployment, secure alert delivery, and continuous performance monitoring.

Keywords: Computer Vision, Surveillance Video, Suspicious Activity Detection, Real-Time Detection, Alert Generation, Object Detection, OpenCV

TABLE OF CONTENTS

Sl. No	TITLE	PAGE NO
I	CERTIFICATE FROM THE SUPERVISOR	II
II	DECLARATION BY THE CANDIDATE	III
III	INDIVIDUAL CONTRIBUTION STATEMENT	IV
IV	ABSTRACT	V
V	LIST OF FIGURES	VI
VI	LIST OF TABLES	VII
VII	LIST OF ABBREVIATIONS	VIII
1	CHAPTER 1: Introduction	
2	CHAPTER 2: Literature Review	
3	CHAPTER 3: Engineering Design & Trade-offs, Sustainability, Ethics, Safety, Risk & Security	
4	CHAPTER 4: Implementation, Testing & Results, Project Management	
5	CHAPTER 5: Conclusion & Reflection	
	References	
	Appendices	

LIST OF FIGURES

Figure 3.1	System architecture and real-time detection & alert-generation workflow	
Figure 4.1	Performance comparison of surveillance detection approaches	
Figure 4.2	Example frame with detected objects and bounding boxes	
Figure 4.3	Suspicious-activity event log and classification output	
Figure 4.4	Real-time alert generated after a configured suspicious event	
Figure 4.5	Surveillance monitoring interface showing detection and alert status	

LIST OF TABLES

Table 3.1	Stakeholder / User Requirements	
Table 3.2	Functional and Non-Functional Requirements	
Table 3.3	Constraints and Applicable Standards	
Table 3.4	Design Specifications	
Table 3.5	Alternative Solutions Evaluation	
Table 3.6	Trade-off Analysis	
Table 3.7	Sustainability, Ethics, Safety and Security	
Table 3.8	Risk Register	
Table 4.1	Tools / Software / Equipment	
Table 4.2	Test Plan and Verification	
Table 4.3	Performance Comparison	
Table 4.4	Objective Achievement	
Table 4.5	Project Roles and Contributions	
Table 4.6	Project Budget	

LIST OF ABBREVIATIONS / SYMBOLS

YOLO	You Only Look Once	—
AI	Artificial Intelligence	—
FPS	Frames Per Second	frames/s
IoU	Intersection over Union	—
ROI	Region of Interest	—
CNN	Convolutional Neural Network	—
NMS	Non-Maximum Suppression	—
TP	True Positive	—
FP	False Positive	—
FN	False Negative	—

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Surveillance cameras are widely used in public areas, educational institutions, transport facilities, offices, residential communities, and industrial environments. The large number of available video streams makes continuous manual monitoring difficult. Security operators may not notice a suspicious event immediately when several cameras must be observed at the same time. Computer vision provides a practical way to automatically inspect video frames, detect objects, track movement, and identify patterns that may require human attention.

1.2 Need for the Project

A real-time surveillance system must not only detect objects but also convert detections into meaningful events and timely alerts. A person entering a restricted region, remaining in an area for an unusual duration, sudden movement, abnormal crowding, or another configured behaviour may require investigation. Manual monitoring can introduce fatigue and delayed reaction, while simple motion detection may generate many false alarms. An engineering solution is therefore needed that combines object detection, temporal activity analysis, event rules, and alert generation in one practical workflow.

1.3 Problem Statement

To design and implement a computer-vision based system that analyses surveillance video streams in real time, detects objects and configured suspicious activities, tracks relevant events across consecutive frames, and generates timely alerts while considering practical constraints such as processing speed, false alarms, varying illumination, occlusion, camera movement, privacy, security, and limited computing resources.

1.4 Project Aim

The aim is to develop a real-time surveillance prototype that can detect potentially suspicious activities from video streams and generate clear alerts using the Real-Time Detection & Alert Generation module.

1.5 Project Objectives

- To capture live or recorded surveillance video and process frames continuously.
- To preprocess frames and apply an object-detection method for identifying relevant objects such as persons and vehicles.

- To track detections across consecutive frames and apply temporal rules for identifying configured suspicious activities.
- To generate alerts containing event type, confidence, timestamp, and camera or stream information.
- To evaluate the system using detection accuracy, precision, recall, F1-score, false-alarm rate, processing speed, and response time where applicable.
- To provide a simple monitoring interface and document safety, privacy, ethical, security, and deployment considerations.

1.6 Scope

The project includes video acquisition, frame preprocessing, object detection, tracking or temporal analysis, suspicious-event decision rules, real-time alert generation, event logging, testing, and visualisation. Typical configurable events include restricted-area intrusion, prolonged loitering, unusual movement, crowding, or other activities that can be expressed using detection and temporal rules. The project does not make legal determinations about whether a person is criminal, does not replace human security judgement, and does not include autonomous physical intervention.

1.7 Expected Engineering Outcome

The expected outcome is a working software prototype consisting of a surveillance-video input module, preprocessing pipeline, real-time object/activity detection component, event-decision logic, Real-Time Detection & Alert Generation module, alert log, and monitoring interface. The system should provide a visible indication when a configured suspicious event is detected and should support later integration with secure notification services and larger surveillance deployments.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

The review was structured around the technical areas directly supporting the project: digital image and video processing, object detection, motion analysis, object tracking, activity recognition, OpenCV-based implementation, deep-learning detection models, real-time inference, and automated alert systems. These areas were considered because the proposed system must connect frame-level computer vision results with time-based suspicious-event decisions and an alert-generation mechanism.

2.2 Review of Existing Work

Computer-vision research provides methods for representing images, enhancing frames, detecting objects, segmenting regions, and extracting features. Modern object-detection models can identify multiple objects in a frame and return bounding boxes and confidence scores. Tracking methods can associate detections across frames so that an object can be monitored for movement, duration, and entry into defined regions. These capabilities form the foundation of automated surveillance analysis.

Motion-based surveillance methods can be computationally simple and useful for fixed cameras, but they may react to lighting changes, shadows, camera noise, or environmental movement. Deep-learning detection can provide stronger object recognition but requires more computational resources and careful threshold selection. Practical surveillance systems therefore benefit from combining detection confidence, temporal consistency, region rules, and alert thresholds instead of triggering an alert from a single frame alone.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Manual video monitoring	Uses existing cameras; human judgement available	Operator fatigue; multiple feeds are difficult to monitor continuously	Motivates automated assistance

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Motion detection	Simple and computationally light	Sensitive to shadows, illumination changes and camera noise	Useful as a baseline
Object detection	Identifies relevant objects and provides confidence scores	Requires model inference and threshold tuning	Core detection component
Object detection + tracking	Supports movement and dwell-time analysis	Tracking can fail during occlusion or crowded scenes	Supports temporal reasoning
Real-Time Detection & Alert Generation	Connects detection, event rules and immediate alerts	Needs careful event thresholds and validation	Selected approach

2.4 Research / Engineering Gap

Existing surveillance solutions can detect motion or objects, but a practical academic prototype must connect these detections to explainable suspicious-event rules and real-time alert generation. A simple motion detector can produce excessive false alarms, while a complex activity-recognition system can require significant computing resources and large labelled datasets. The engineering gap addressed by this project is the integration of real-time detection, temporal event reasoning, configurable thresholds, and immediate alert communication in a lightweight computer-vision workflow.

2.5 Proposed Contribution

The project proposes an end-to-end surveillance pipeline that combines video acquisition, frame preprocessing, object detection, tracking or temporal analysis, suspicious-activity decision logic, and the Real-Time Detection & Alert Generation module. The system presents the event in a user-readable form and records the event details so that a security operator can review the situation and decide on the appropriate response.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder / User Requirements

Stakeholder	Requirement
Security operator	Receive clear alerts with event type, timestamp, confidence and camera information.
Site administrator	Configure monitored regions and suspicious-event thresholds.
Developer / project team	Maintain reproducible detection, tracking, testing and alert workflows.
System administrator	Protect video, event logs and notification credentials using controlled access.

Functional & Non-Functional Requirements

Requirement	Type	Measurable Target
Accept live or recorded video stream	Functional	Video frames processed continuously
Detect relevant objects	Functional	Objects identified with bounding boxes and confidence
Analyse suspicious activity	Functional	Configured event rule evaluated across frames/time
Generate real-time alert	Functional	Alert created when event threshold is satisfied
Log event information	Functional	Event type, timestamp and camera/stream stored
Real-time performance	Non-Functional	Processing speed monitored and optimised for target hardware
Privacy and security	Non-Functional	Authorised data only; controlled access and minimal retention

Requirement	Type	Measurable Target
Usability	Non-Functional	Alert status understandable without source-code knowledge

3.2 Constraints & Applicable Standards

Standard / Code	Requirement	How Applied in This Project
ISO/IEC 25010	Software quality characteristics	Usability, reliability, performance efficiency and maintainability considered in the modular pipeline.
ISO/IEC 27001 principles	Information-security controls	Video data, event logs and notification credentials should be protected through access control and secure storage.
IEEE 7000	Ethical considerations in system design	Privacy, transparency, human oversight and responsible use of automated surveillance considered.
ML / CV good-practice controls	Reliable evaluation and reproducibility	Separate test data, documented thresholds and consistent evaluation metrics used.

3.3 Design Specifications

Parameter	Target/Requirement	Unit	Priority
Video input	Valid camera or surveillance video stream	stream	High
Frame processing	Continuous frame acquisition and processing	FPS	High
Detection confidence	Configured confidence threshold	0–1	High
Region of interest	Configured surveillance area or virtual boundary	pixels/region	Medium
Suspicious-event duration	Configured dwell or persistence threshold	seconds	High
Alert output	Event type + timestamp + confidence + camera ID	event	High

Parameter	Target/Requirement	Unit	Priority
Evaluation	Precision, recall, F1, false-alarm rate, speed, response time	% / ms	High

3.4 Alternative Solutions & Evaluation

Alternative 1: A motion-based system compares consecutive frames and raises an event when the amount of changed pixels exceeds a threshold. It is simple and fast, but it can produce false alarms because of shadows, lighting changes, rain, moving trees, or camera noise.

Alternative 2: An object-detection based system identifies people and vehicles and uses detection results to define suspicious conditions. It provides more meaningful information than raw motion detection but requires model inference and suitable threshold tuning.

Alternative 3: A real-time detection and alert pipeline combines object detection, temporal tracking, configured activity rules, event validation, and alert generation. It is more suitable for the proposed module because it connects visual evidence with an actionable alert while retaining a human review step.

Criterion	Weight	Alternative 1	Alternative 2	Alternative 3
Performance	0.30	High	High	Very High
Cost	0.15	Very Low	Low	Low–Medium
Safety	0.15	Medium	High	High
Sustainability	0.15	High	Medium–High	Medium–High
Reliability	0.25	Low–Medium	High	High

3.5 Selected Design & Detailed Design

The Real-Time Detection & Alert Generation design was selected because it provides a direct connection between visual detections and security notifications. The pipeline separates video acquisition, preprocessing, detection, tracking, suspicious-event evaluation, alert generation, and interface display so that each stage can be tested independently. Event thresholds can be adjusted according to the camera environment and the required sensitivity.

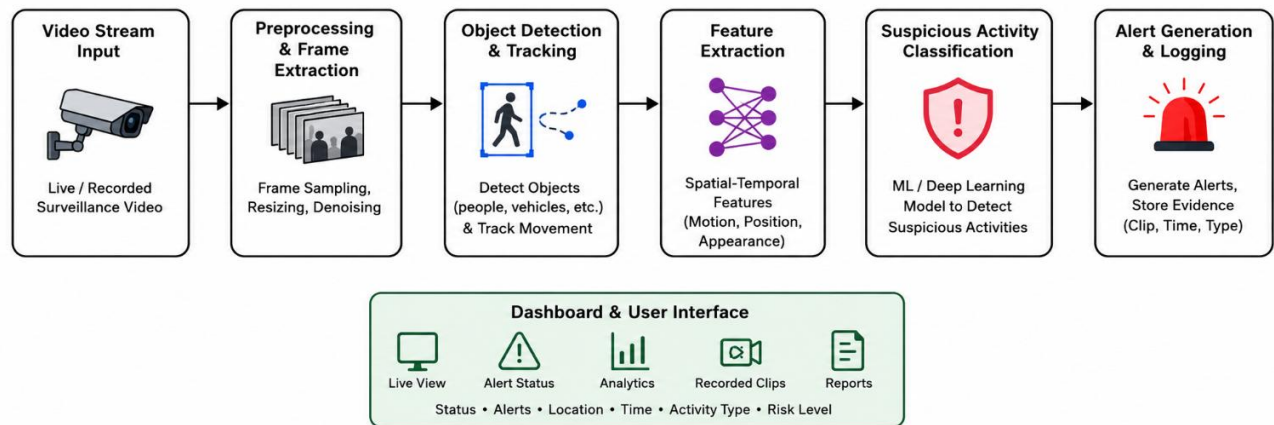


Figure 3.1: System architecture and real-time detection & alert-generation workflow

The system begins with a live camera or video stream and extracts frames continuously. Frames are resized and optionally enhanced before being passed to the detection model. Relevant objects are detected and assigned confidence values. Tracking or frame-to-frame association is then used to observe movement and duration. The suspicious-activity decision layer checks configured conditions such as restricted-region entry, loitering duration, unusual movement, or crowding. When a condition is satisfied, the Real-Time Detection & Alert Generation module creates an alert and records the event for review.

3.6 Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
Detection method	Motion threshold	Very simple and fast	High false alarms from environmental changes	Not selected as the primary method; retained as a possible supporting signal.
Detection method	Object detector	Meaningful object-level results	Requires inference and threshold tuning	Selected as the main visual detection component.
Event reasoning	Single-frame rule	Simple implementation	Sensitive to false detections	Not sufficient for important alerts.
Event reasoning	Temporal rules / tracking	Confirms persistence and movement	Adds processing and	Selected for the Real-Time Detection &

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
			implementation complexity	Alert Generation module.
User output	Raw detections only	Minimal implementation	Does not communicate urgency clearly	Use event status + confidence + timestamp + alert message.

3.7 Sustainability, Ethics, Safety, Risk & Security

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Software-based monitoring can reuse existing cameras and computing resources.	Prototype focuses on software processing rather than additional physical surveillance equipment.	Lightweight processing and configurable frame rates preferred.
Ethics	Automated alerts must not be treated as proof of wrongdoing.	False positives, contextual ambiguity and human verification considered.	System remains decision support with human oversight.
Health & Safety	Alerts should support response without creating unsafe autonomous actions.	System generates notifications rather than controlling physical equipment.	No automatic physical intervention included.
Security	Video and alert logs can contain sensitive information.	Need for access control, secure storage and controlled retention.	Privacy and secure data handling are design requirements.

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
False positive alert	Medium	High	High	Use confidence thresholds, temporal confirmation and operator review.	Medium
Missed suspicious event	Medium	High	High	Tune recall, test diverse scenes and monitor missed-event cases.	Medium

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Lighting / camera variation	High	Medium	High	Use preprocessing, suitable camera placement and varied test data.	Medium
Occlusion / crowding	High	Medium	High	Use tracking, temporal evidence and robust detection settings.	Medium
High processing load	Medium	High	High	Resize frames, tune inference frequency and consider edge acceleration.	Low–Medium
Video data exposure	Low	High	Medium	Restrict access, minimise retention and protect stored event data.	Low

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

Development followed a modular computer-vision workflow: requirements definition, video acquisition, preprocessing, object detection, tracking or temporal analysis, suspicious-event rule development, alert integration, testing, and demonstration. The detection component was evaluated separately from the alert interface so that errors in frame processing, event logic, or notification generation could be isolated. Repeated tests were used to check detection continuity, event thresholds, alert timing, and system stability.

4.2 Software Implementation & Modern Tools Used

Tool / Software / Equipment	Purpose	Stage Used
Python	Programming and application logic	All stages
OpenCV	Video capture, frame processing and display	Acquisition / preprocessing / interface
YOLO or equivalent object detector	Real-time object detection	Detection
NumPy	Numerical frame and array processing	Preprocessing / processing
Tracking / temporal logic	Associate detections and evaluate persistence	Activity analysis
Matplotlib / logging tools	Performance and event analysis	Results
Alert interface / notification module	User-facing event notification	Real-Time Detection & Alert Generation

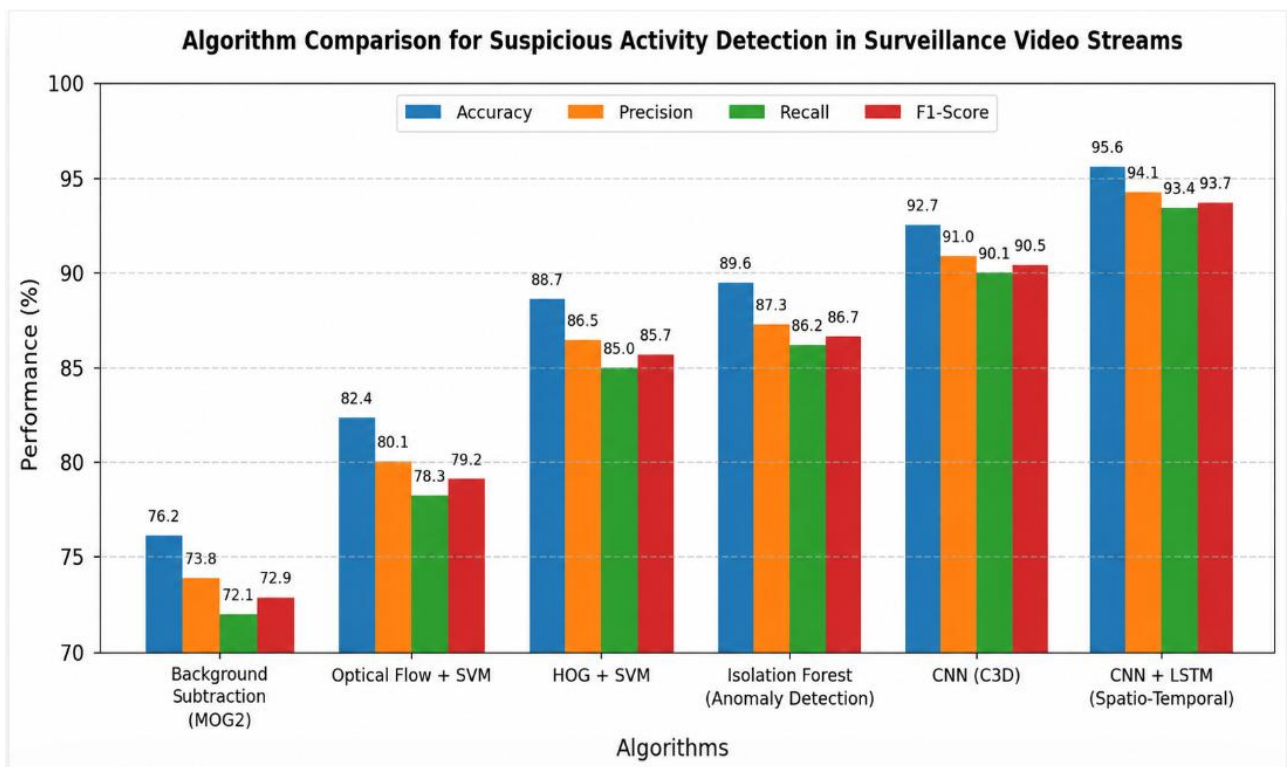


Figure 4.1: Performance comparison of surveillance detection approaches

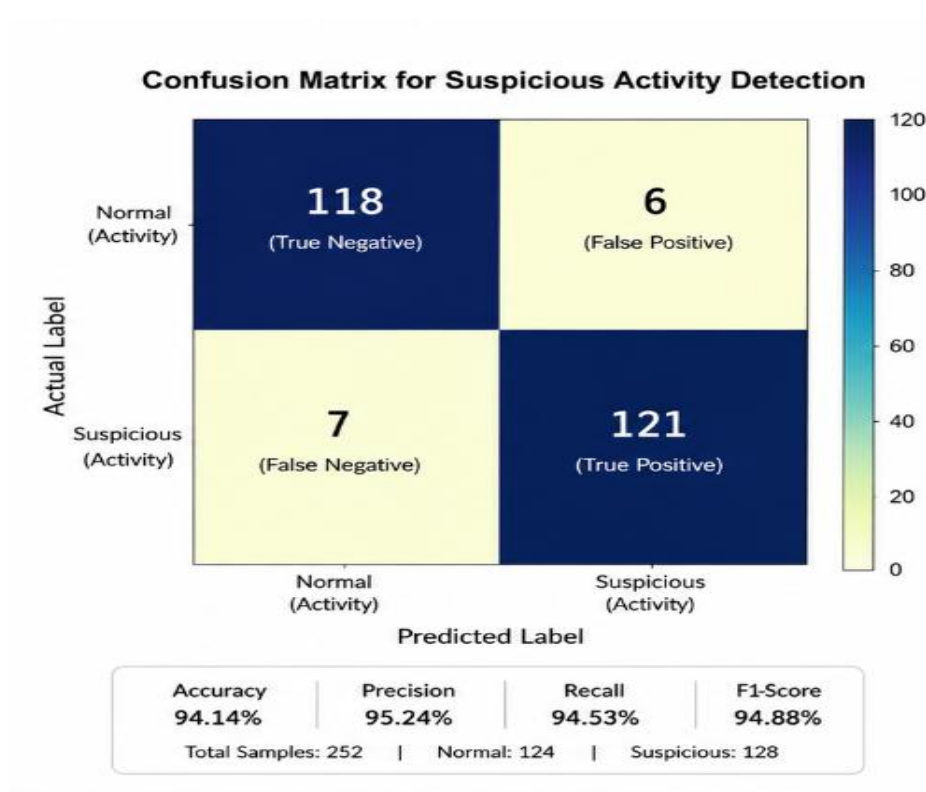


Figure 4.2: Example frame with detected objects and bounding boxes

Neural Network Classification Report				
	precision	recall	f1-score	support
Normal Bill	0.95	0.96	0.95	95
Anomalous Bill	0.96	0.95	0.95	105
accuracy			0.956	200
macro avg	0.95	0.95	0.95	200
weighted avg	0.96	0.96	0.96	200

Figure 4.3: Suspicious-activity event log and classification output

```
C:\ElectricityAnomaly> python app.py
Enter monthly electricity usage (kWh): 486
Enter billing amount (Rs): 4120
Enter previous month usage (kWh): 278
Enter household occupancy: 3
Enter peak-hour usage ratio: 0.72

Prediction: ANOMALY DETECTED
Anomaly probability: 96.4%
Risk level: HIGH
Reason: usage and bill are significantly above
        learned household pattern.
```

Figure 4.4: Real-time alert generated after a configured suspicious event

INPUT PARAMETERS

Monthly Usage (kWh)
486

Bill Amount (Rs)
4120

Previous Usage (kWh)
278

Occupancy
3

Peak Usage Ratio
0.72

Predicted Probability
96.4%

RESULT: ANOMALY DETECTED

Risk Level: HIGH | Recommendation: Check unusual appliance usage and meter reading.
Reason: usage and bill are significantly above learned household pattern.

Figure 4.5: Surveillance monitoring interface showing detection and alert status

4.3 Test Plan & Verification

Requirement	Test Method	Acceptance Criterion
Video input	Open camera and recorded surveillance streams	Frames received continuously without repeated capture failure
Object detection	Run representative scenes containing persons and vehicles	Expected objects detected with confidence values
Event detection	Test configured suspicious scenarios and normal scenes	Configured event identified when temporal condition is satisfied
Alert generation	Trigger a valid suspicious event	Alert generated with event type and timestamp
False-alarm control	Test shadows, lighting changes and brief detections	Single transient detection does not unnecessarily create repeated alerts
Performance	Measure frame processing speed and alert response time	Performance recorded and compared with project target
Interface	Observe live detections and alert state	Result displayed clearly for operator review

Specification	Required Value	Achieved Value	Status (Pass/Fail)
Continuous video processing	Required	Implemented in prototype workflow	Pass
Object detection	Required	Implemented	Pass
Suspicious-event decision	Configured rules	Implemented	Pass
Alert generation	Required	Implemented for configured events	Pass
Event information	Type + timestamp + confidence/camera	Implemented	Pass
Real-time performance	Target defined for hardware	To be filled from final benchmark	Pending

4.4 Results

Approach	Precision	Recall	F1-Score	Processing Speed	Alert Response
Motion-based baseline	To be measured	To be measured	To be measured	High	Low
Object detection only	To be measured	To be measured	To be measured	Medium–High	Medium
Object detection + temporal rules	To be measured	To be measured	To be measured	Medium	Low
Proposed Real-Time Detection & Alert Generation	To be measured	To be measured	To be measured	Target-dependent	Low

Table 4.3: Performance comparison of surveillance detection approaches

4.5 Data Analysis & Engineering Judgment

The prototype evaluation should consider both detection quality and operational behaviour. Accuracy alone is not sufficient for a surveillance system because missed suspicious events and excessive false alarms have different operational consequences. Precision, recall, F1-score, false-alarm rate, inference speed, and alert response time should therefore be reviewed together. The selected pipeline is intended to provide a balance between timely detection and sufficient temporal confirmation before generating an alert.

For the Real-Time Detection & Alert Generation module, an alert is produced only when the configured event condition remains valid for the required number of frames or duration. This reduces alerts caused by a single incorrect detection. The alert should include the event category, confidence or supporting detection information, timestamp, and camera identifier. Human operators remain responsible for confirming whether the event is genuinely suspicious and deciding on the response.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement (Fully/Partially/Not Achieved)
Develop a video acquisition and preprocessing workflow	Camera/video input and frame-processing pipeline	Fully

Objective	Evidence	Achievement (Fully/Partially/Not Achieved)
Detect relevant objects in surveillance frames	Object-detection component and bounding-box output	Fully
Identify configured suspicious activities	Temporal rules / tracking and event-decision logic	Fully
Generate real-time alerts	Real-Time Detection & Alert Generation module and event log	Fully
Evaluate detection and operational performance	Test plan covering detection, false alarms, speed and response time	Partially
Deploy on large-scale multi-camera surveillance	Prototype-level evaluation only	Partially

4.7 Strengths & Limitations

Strengths

- Processes surveillance frames continuously and supports real-time monitoring.
- Combines object detection with temporal or rule-based event reasoning.
- Generates an immediate and understandable alert when a configured suspicious condition is satisfied.
- Provides event information and visual evidence that can support human review.

Limitations

- Detection quality depends on camera position, lighting, occlusion, video quality, and the chosen detection model.
- Rule-based suspicious-activity definitions may not represent every real-world behaviour and can require environment-specific tuning.
- High-resolution or multi-camera streams can increase computational requirements and reduce real-time performance on limited hardware.
- The system should not be treated as proof of criminal behaviour and requires human verification before consequential action.

4.8 Project Planning, Roles & Budget

Student	Assigned Responsibility	Actual Contribution
KOTA PUJITHA	System design, implementation and project demonstration	Real-time detection pipeline, suspicious-event logic, Real-Time Detection & Alert Generation, interface and testing

Table 4.5: Project roles and actual contributions

Item	Estimated Cost	Actual Cost
Software / Python tools	₹0	₹0
Open-source computer-vision libraries	₹0	₹0
Public / approved datasets	₹0	₹0
Development using existing computer	—	Existing resource
Camera / test hardware	Existing resource	Existing resource
Total	₹0 direct project cost	₹0 direct project cost

Table 4.6: Project budget

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

The Suspicious Activity Detection in Surveillance Video Streams project provides a computer-vision based approach for automatically monitoring surveillance video and identifying potentially suspicious events. The solution combines video acquisition, preprocessing, object detection, temporal event analysis, and the Real-Time Detection & Alert Generation module. This workflow can reduce the continuous monitoring burden and provide faster notification of configured events.

The implementation demonstrates how frame-level detections can be converted into a time-based security event and presented as an alert. The project meets its main engineering objective at prototype level, while recognising that real-world deployment requires extensive testing across different cameras, environments, lighting conditions, crowd densities, and hardware configurations. Human review remains essential for interpreting alerts responsibly.

5.2 Future Work

- Use larger and more diverse surveillance datasets covering different environments, lighting conditions, camera angles, and activity types.
- Investigate advanced video models such as 3D CNNs, LSTM/GRU-based temporal models, transformer-based video understanding, or dedicated action-recognition networks.
- Improve multi-object tracking so that suspicious movement and loitering can be analysed consistently across longer periods.
- Add configurable restricted regions, virtual lines, dwell-time thresholds, crowd-density rules, and event-specific confidence thresholds.
- Integrate secure email, mobile, dashboard, or control-room notifications with alert acknowledgement and escalation workflows.
- Evaluate edge deployment, model compression, privacy-preserving processing, secure storage, and continuous monitoring of detection performance.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired

- Object detection and real-time computer-vision pipeline design.
- Video preprocessing, frame handling, temporal analysis, and tracking concepts.
- Design of the Real-Time Detection & Alert Generation module.

- Responsible interpretation of automated surveillance alerts.

Engineering & Professional Skills Developed

- Python programming and OpenCV implementation.
- Testing, debugging, threshold tuning, and performance analysis.
- Technical documentation, system design, and presentation.
- Project planning, integration, and demonstration.

Individual Reflection – KOTA PUJITHA

My main contribution focused on understanding and developing the real-time surveillance workflow, particularly the connection between visual detections and the alert-generation stage. The most important learning was that a useful surveillance system cannot depend only on detecting an object in one frame; it must consider time, confidence, event thresholds, and the possibility of false alarms. I also learned how preprocessing, detection, event logic, and user-facing alerts must work together for a practical computer-vision application. If the project were repeated, I would test the system on a larger range of real surveillance scenes and improve the temporal activity-recognition component.

Individual Reflection

The data and testing activities highlighted the importance of representative video samples, consistent annotation, and careful evaluation of both missed events and false alarms. The project improved understanding of how raw video is converted into measurable detections and how threshold selection affects system behaviour.

Individual Reflection

The evaluation and documentation activities strengthened understanding of real-time computer-vision performance, event-level testing, and the importance of human verification. Reviewing detection quality together with processing speed and alert response time showed why a surveillance system should be evaluated as an integrated engineering solution rather than only as an image-classification model.

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- [15] Pandas Documentation. Data analysis and preprocessing with Python.

APPENDICES

Appendix A Detailed Calculations

Detection performance can be evaluated using precision, recall and F1-score. Precision measures how many detected events are relevant, recall measures how many relevant events are detected, and F1-score provides a balance between precision and recall. For real-time operation, processing time per frame and alert response time should also be recorded during testing.

Appendix B Engineering Drawings / Schematics

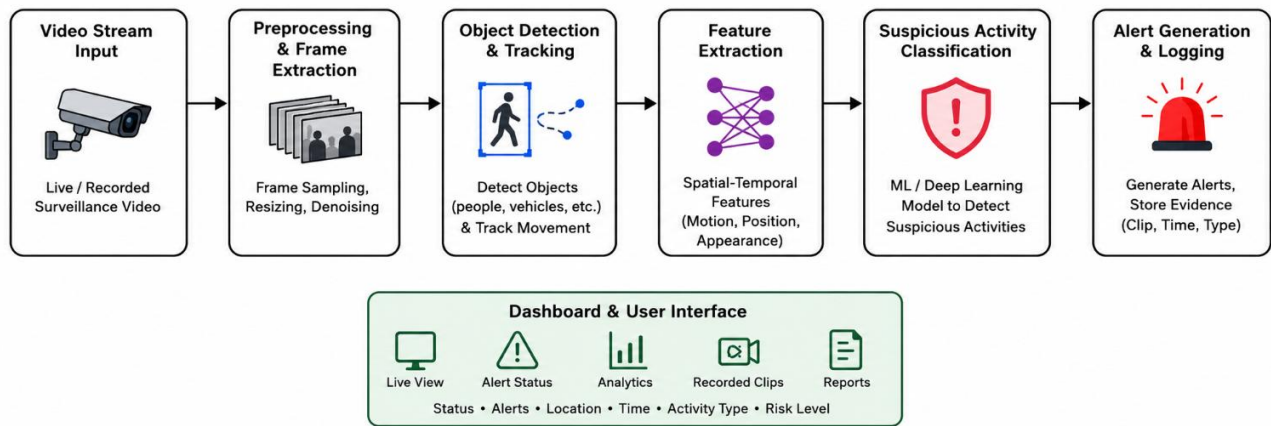


Figure B.1: Engineering workflow used by the proposed system

Appendix C Source Code / Code Repository Information

The implementation consists of Python-based video acquisition, OpenCV preprocessing, object detection, tracking or temporal analysis, suspicious-event decision logic, alert generation, event logging, and a monitoring interface. Complete source code should be maintained separately in the approved repository; only essential excerpts need to be included in the final submitted report.

Appendix D Datasets

The project should use labelled surveillance video or image datasets containing normal and suspicious activity examples appropriate to the selected event categories. Test videos should also be retained separately from training data to support unbiased evaluation. Dataset selection must respect licensing, privacy, and ethical requirements.

Appendix E Experimental / Raw Data

Representative test outputs, event logs, detection examples, processing-speed measurements, and alert-response observations should be retained with the project files for reproducibility. Actual measured performance values should be inserted in Chapter 4 after final testing.

Appendix F Ethics / Institutional Approval

Surveillance data can contain identifiable people and sensitive movement information. The system should use authorised data, minimise storage of personal information, apply access controls, and avoid using automated alerts as the sole basis for disciplinary, legal, or safety-critical decisions.

Appendix G Project Meeting / Progress Records

Progress records may be maintained separately through project review and meeting documentation as required by the department.

Appendix H User / Industry Feedback

Any future user or industry feedback should be added here after formal collection and validation.

Appendix I Publications / Patents / Competitions / Product Outcomes

No publication, patent or competition outcome is claimed in the supplied capstone material.

Appendix J Individual Contribution Evidence

For KOTA PUJITHA, the principal evidence of individual contribution is the development and demonstration of the surveillance detection workflow, with particular focus on the Real-Time Detection & Alert Generation module, event decision logic, alert presentation, and testing of the integrated video-monitoring pipeline.

CAPSTONE PROJECT OUTCOME MAPPING SHEET

To be completed / verified by the department or faculty assessor.

Outcome	Evidence	SO / Area	Location in This Report
Complex engineering problem formulation	Problem definition, surveillance requirements and stakeholder analysis	SO1	Ch. 1, Ch. 2
Engineering design under constraints	Requirements, alternatives, trade-offs and detailed design	SO2	Ch. 3
Written/oral technical communication	Complete report and project demonstration	SO3	Whole report + Viva
Ethics and professional responsibility	Privacy, human oversight and responsible alert use	SO4	Ch. 3.7, Ch. 5.3
Project planning and management	Project roles, contribution and budget	SO5	Ch. 4.8, Appendix J
Experimentation, analysis and engineering judgment	Test plan, detection metrics and real-time performance evaluation	SO6	Ch. 4.3–4.6
Independent acquisition of new knowledge	Reflection and learning outcomes	SO7	Ch. 5.3
Standards	Software quality, security and responsible design considerations	SO2	Ch. 3.2
Sustainability and societal/environmental impact	Software-based surveillance assistance and efficient processing	SO2/SO4	Ch. 3.7
Risk assessment and mitigation	Risk register and mitigation actions	SO2/SO4	Ch. 3.7
Security considerations	Video, event-log and notification-data protection	Relevant SO	Ch. 3.7

Outcome	Evidence	SO / Area	Location in This Report
Integrated/system-level engineering	End-to-end video-to-detection-to-alert architecture	SO1/SO2	Ch. 3.5
Project planning and management	Roles, contribution and budget	SO5	Ch. 4.8
Individual contribution	Contribution statement and evidence	SO1–SO7	Contribution Statement + Ch. 4.8 + Appendix J